



Technical Service Bulletin

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New Valve Guides

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Warranty Statement

The information in this document has no effect on present warranty coverage or repair practices, nor does it authorize TRP or Campaign actions.

Contents

This document announces the release of new valve guides for the cylinder heads installed on the QSK19 CM500 HPI, QSK45/60 CM500 HPI, QSK60 CM850 MCRS, and QSK60 CM2150 MCRS engines.

QSK19 CM850 MCRS, QSK19 CM2150 MCRS, QSK19 Gas, and QSK60 Gas engines are **not** affected by this change to the valve guide.

The new valve guide has a revised internal diameter tolerance to make sure correct valve stem to valve guide clearance is maintained. The new valve guide is also available with two stages of oversize outer diameter for installation in heads where the valve guide bore in the cylinder head has been machined.

Valve guide						
Description	Outside Diameter				Old Part Number	New Part Number
	mm		Inch			
	Minimum	Maximum	Minimum	Maximum		
Valve Guide: Standard	21.473	21.485	0.8454	0.8459	3641624	3646200
Valve Guide: Oversize 1	21.727	21.739	0.8554	0.8559	3641625	3646201
Valve Guide: Oversize 2	21.854	21.866	0.8604	0.8609	3641626	3646202

Due to the introduction of the new valve guides, the part numbers of the cylinder head assemblies have also been superseded.

Cylinder Heads				
Cylinder Head Assembly		Intake Valve Specified		Application
Old Part Number	New Part Number	Old Part Number	New Part Number	
3641569	3646783	3641538	3641538	QSK19 CM500 HPI
3641560	3647147	3641540	3641540	QSK19 CM500 HPI
3641567	3646784	3641540	3641540	QSK60 Tier 2 MCRS - Industrial and Power Generator (Non-Marine certified) No longer used - use Part Number 3646785.
3645552	3646785	3641538	3641538	QSK60 Tier 2 MCRS - Industrial and Power Generator (Non-Marine certified)
3641561	3647148	3641539	3641539	QSK60 CM500 HPI Power Generator (PP8613 and PP8617 only)
3641565	4313888	3641540	3641538	QSK60 CM500 HPI - Marine certified
3641558	4313889	3641540	3641538	QSK60 Tier 2 MCRS - Marine certified
3641562	3647149	3641538	3641538	QSK45/60 CM500 HPI Industrial and Power Generator (Non-Marine certified)

Cylinder Heads				
Cylinder Head Assembly		Intake Valve Specified		Application
3641564	3647150	3641540	3641538	QSK60 CM500 HPI – Marine, but not certified. No longer used - use Part Number 3647149.

Note : Included in the table are changes to the intake valves specified for cylinder heads utilized in Marine applications, and to those used on QSK60 Tier 2 MCRS (CM850 and CM2150) engines. The intake valves specified have changed from a 'K' marked valve to a 'C' marked valve, in order to bring these cylinder heads in-line with other industrial cylinder heads.

Due to the revision to the valve guide tolerance, it is necessary to employ a new installation procedure when installing valve guides into cylinder heads.

A drop gauge test has always been required following valve guide installation into the cylinder head. A new drop gauge is required to make sure that the improved valve guide internal diameter tolerance is maintained following installation of the guide into the cylinder head. The new drop gauge has an increased diameter.

A new reamer has also been released, to avoid the requirement to remove and replace valve guides due to the drop gauge test **not** passing. The reamer is **only** to be employed following an unsuccessful drop gauge test.

When a valve guide is pressed into the cylinder head, some slight distortion to the internal diameter of the valve guide can occur. The purpose of the reamer is to remove a very small amount of material from any areas of distortion in the internal diameter of the installed valve guide. This makes sure the correct internal diameter is maintained and allows the drop gauge test to be successful.

Required Items		
	Old Part Number	New Part Number
Drop Gauge (Figure 1)	4918779	5298508
Reamer (Figure 2)	N/A	5298510

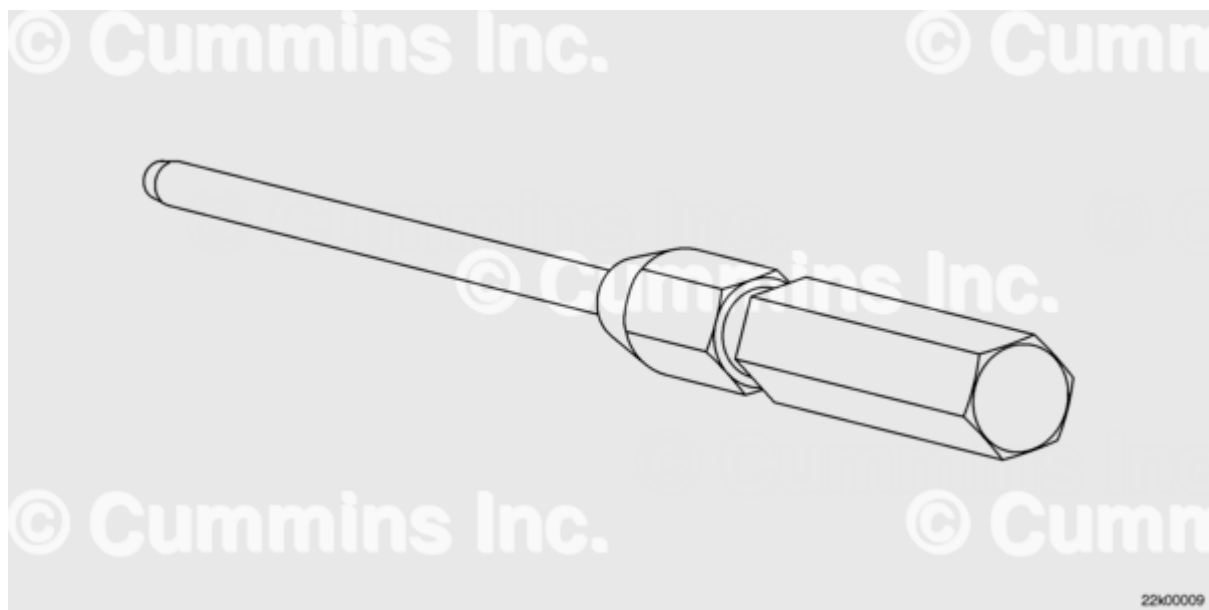


Figure 1

Drop Gauge, Part Number 528508.

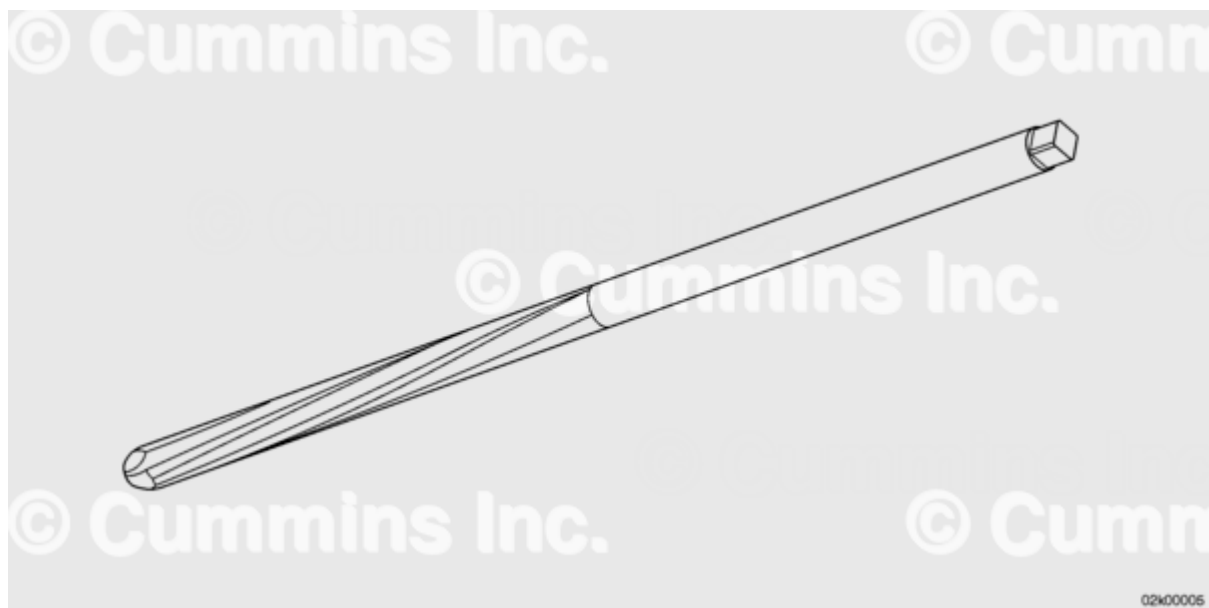


Figure 2

Reamer, Part Number 5298510.

Reference the appropriate procedure below:

- Use the following procedure in the QSK19, QSK19 CM850 MCRS, and QSK19 CM2150 MCRS Service Manual, Bulletin 4021592. Refer to Procedure 002-004 in Section 2. (</qs3/pubsys2/xml/en/procedures/20/20-002-004-tr.html>)
- Use the following procedure in the QSK45 and QSK60 Service Manual, Bulletin 4021530. Refer to Procedure 002-004 in Section 2. (</qs3/pubsys2/xml/en/procedures/56/56-002-004-tr.html>)

Document History

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2012-4-19	Module Created

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